

Module 02: Agents in High School

Learning Objectives

1. Define **Agents** within the context of High School.
2. Analyze how Agents interacts with other components of the Active Inference framework.
3. Apply specific constraints of High School to the formal definition of Agents.

Introduction

This module explores **Agents**. In the **High School** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Agents is a critical component of the 8-part Active Inference spine, bridging the gap between Systems and Perception.

Key Concepts

1. Agents as a Markov Blanket Boundary

How does Agents define the boundary between the agent and the environment?

2. Generative Models of Agents

What parameters involved in Agents must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Agents drive the perception-action loop?

Applications

In High School, we see Agents manifest in: * **Specific Example 1:** In game theory, a chess-playing agent can be modeled mathematically as an Active Inference system: the agent

maintains a probability distribution over possible board states (beliefs), updates these beliefs after each opponent move using Bayes' theorem, and selects the move that minimizes expected free energy -- balancing exploitation of known good positions with exploration of uncertain but potentially advantageous lines. * **Specific Example 2:** A simple random walk on a number line (where a point moves +1 or -1 with equal probability) becomes an Active Inference agent when you add a preferred state: if the agent "wants" to stay near zero, it biases its step probabilities using a mathematical objective function that penalizes deviation, turning a passive stochastic process into an agent that actively minimizes surprise relative to its goal.

Conclusion

Understanding Agents allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.